

Candidate Transcriptional Metabolic States in Meningioma scRNA-seq Data

QA_R9 Research Analysis

September 14, 2026

1 Data and Methods

This analysis uses single-cell RNA-seq data from the Choudhury et al. 2022 study (3CA:20773) <https://doi.org/10.1038/s41588-022-01061-8>, comprising 58,843 cells and 33,538 genes from 10 meningioma samples. User-supplied metabolic genes (1,988 symbols in `inputs/metabolic_genes_total`) were matched to the expression matrix via exact-symbol aggregation, yielding 1,646 genes after prevalence filtering.

1.1 Normalization and Feature Scaling

Expression values were normalized using per-cell library size scaling to 10,000 UMI counts, followed by log1p transformation:

$$x'_{g,i} = \log \left(1 + \frac{10^4 \cdot x_{g,i}}{\sum_h x_{h,i}} \right) \quad (1)$$

where $x_{g,i}$ is the raw UMI count for gene g in cell i , and the denominator is the total UMI count across all genes in cell i . Features were then scaled using Scanpy's `scale` function with `zero_center=False` and `max_value=10`.

1.2 Clustering and Baselines

Leiden clustering was performed on the first 30 PCA components (`n_neighbors=15`) at resolution 0.8, selected as the highest median silhouette across five random seeds [0, 17, 42, 73, 101]. Stability was assessed via pairwise ARI across seeds (mean=0.965, min=0.947). A matched-k KMeans control (`k=17`, `seed=0`) yielded silhouette=0.220. Nineteen random-gene controls (size-matched from non-metabolic, non-mitochondrial symbols) produced silhouettes ranging from 0.199 to 0.222, with only 2 of 19 (add-one rank fraction=0.15) achieving silhouette greater than or equal to the metabolic control.

2 Question 1: Do all cells form distinct, reproducible candidate transcriptional metabolic clusters?

Answer: Candidate partitions observed but distinct metabolic states inconclusive.

Global analysis of 58,843 cells using 1,646 matched metabolic genes yielded 17 Leiden clusters at resolution 0.8. Cluster sizes ranged from 714 to 8,806 cells (median approx 3,500). Leiden silhouette=0.217, matched-k KMeans silhouette=0.220, and mean pairwise ARI=0.965 across seeds.

Random-gene controls (19 total) produced silhouettes with mean=0.210, min=0.199, max=0.222. Only 2 controls exceeded or equaled the metabolic control (add-one rank fraction=0.15). The HVG baseline yielded silhouette=0.283, exceeding the metabolic control (0.220).

Interpretation: The observed silhouette (0.217) is comparable to random controls (mean=0.210), and the add-one rank fraction (0.15) indicates that metabolic genes do not systematically outperform random gene sets. The HVG baseline (0.283) is stronger, suggesting that the metabolic gene set does not capture the dominant sources of variation in this dataset. Candidate partitions are descriptive but do not establish distinct, reproducible metabolic states.

3 Question 2: How are those candidate metabolic partitions associated with the supplied cell types and subtypes?

Answer: Associations are descriptive and non-causal.

Categorical confounder NMI values were computed for patient, sample, source, cell_type, cell_subtype, cell_cycle_phase, and complexity. Results show substantial confounding:

- patient: NMI=0.518
- sample: NMI=0.548
- source: NMI=0.191
- cell_type: NMI=0.496
- cell_subtype: NMI=0.648
- cell_cycle_phase: NMI=0.194

Cell-type association summary (global NMI=0.496, replicates_assessable=6) shows within-replicate NMI range [0.379, 0.745], median=0.592. Cell-subtype association (global NMI=0.648, replicates_assessable=6) shows within-replicate NMI range [0.224, 0.680], median=0.636.

Interpretation: High NMI values for patient (0.518) and sample (0.548) indicate that clustering is strongly confounded by sample origin. Cell-type and cell-subtype associations (NMI approx 0.5-0.65) are descriptive and non-causal. These associations cannot establish biological metabolic states, as global cell-type separation does not imply within-cell-type metabolic homogeneity. Patient/sample mixing can confound interpretation.

4 Question 3: Within the exact annotated subset cell_subtype=CD8 T cells, are there distinct candidate metabolic states?

Answer: Candidate partitions observed but distinct metabolic states inconclusive.

Analysis of 3,624 CD8 T cells using 1,318 matched metabolic genes (1,988 source genes to 1,798 before prevalence to 1,318 after) yielded 6 Leiden clusters at resolution 0.4. Cluster sizes ranged from 55 to 2,002 cells. Leiden silhouette=0.099, matched-k KMeans silhouette=0.095, and mean pairwise ARI=0.759 across seeds.

Random-gene controls (19 total) produced silhouettes with mean=0.107, min=0.092, max=0.131. Seventeen of 19 controls (add-one rank fraction=0.9) exceeded or equaled the metabolic control (0.095). The HVG baseline yielded silhouette=0.168, exceeding the metabolic control.

Interpretation: The observed silhouette (0.099) is substantially lower than random controls (mean=0.107), and the add-one rank fraction (0.9) indicates that metabolic genes perform no better than random gene sets. The HVG baseline (0.168) is stronger. Candidate partitions are descriptive but do not establish distinct, reproducible metabolic states within CD8 T cells.

5 Limitations

1. Global cell-type separation cannot establish within-cell-type metabolic states.
2. Patient-within sensitivity is not independent-cohort replication.
3. Patient/sample mixing can confound interpretation.
4. No automatic threshold proves independence, multimodality, mechanism, causality, or statistical significance.
5. Candidate partitions are descriptive; conclusions are inconclusive.

6 Results Summary Table

Table 1: Global and CD8 T cell subset analysis metrics

Analysis	Cells	Clusters	Leiden Silhouette
Global (all cells)	58,843	17	0.217
CD8 T cells	3,624	6	0.099

7 Within-Replicate Sensitivity

Global analysis (patient as replicate field):

- Replicates analyzed: 6
- Global-vs-within ARI range: [0.071, 0.676]
- Weakest replicate: 3 (3,287 cells)

CD8 T cell subset analysis:

- Replicates analyzed: 3
- Global-vs-within ARI range: [0.511, 0.845]
- Weakest replicate: 6 (2,776 cells)

References

- [1] Abrar Choudhury et al. *Meningioma DNA methylation groups identify biological drivers and therapeutic vulnerabilities*. *Nature Genetics* 54(5):649-659, 2022. <https://doi.org/10.1038/s41588-022-01061-8>
- [2] Avishai Gavish et al. *Hallmarks of transcriptional intratumour heterogeneity across a thousand tumours*. *Nature* 618(7965):598-606, 2023. <https://doi.org/10.1038/s41586-023-06130-4>

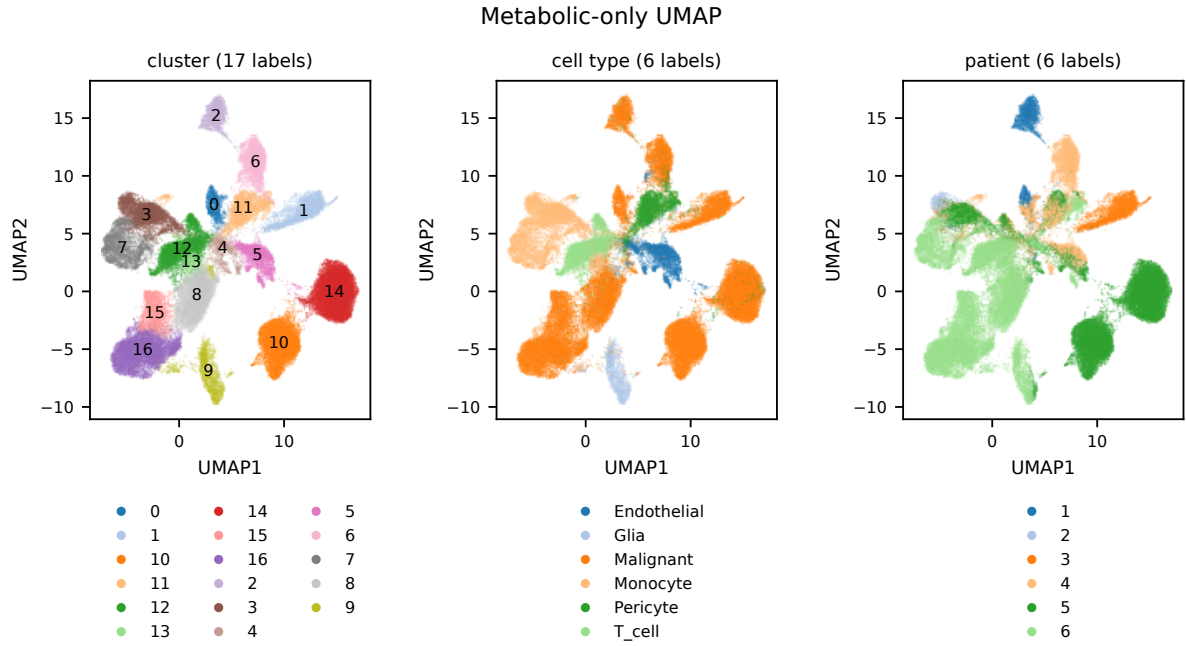


Figure 1: Global metabolic UMAP of 58,843 cells clustered on 1,646 metabolic genes. 17 Leiden clusters at resolution 0.8.

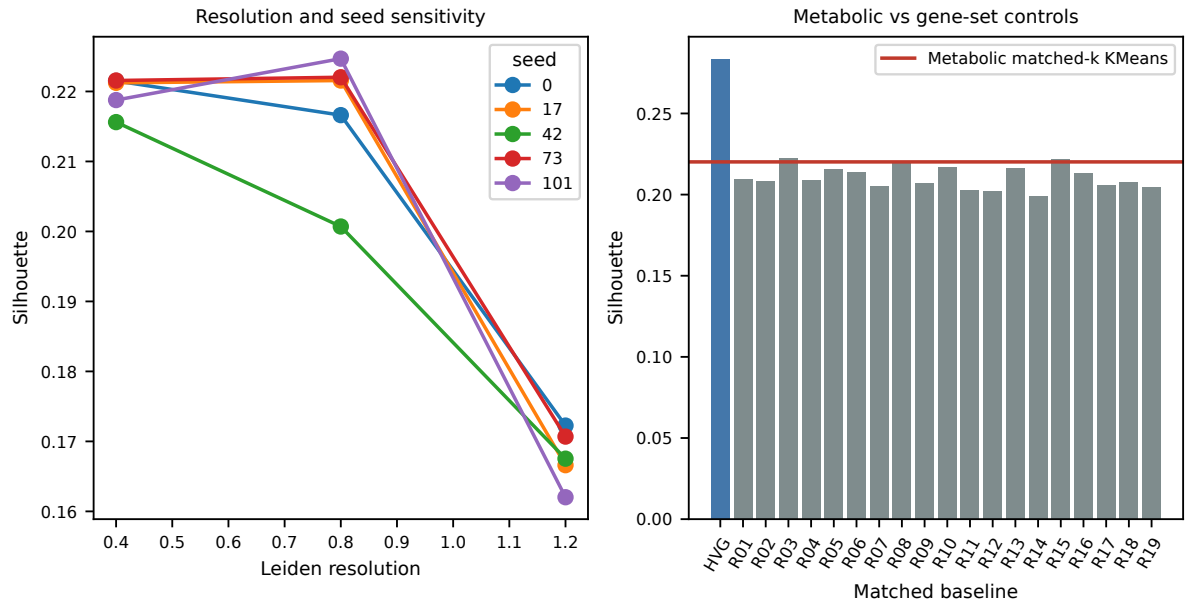


Figure 2: Global clustering diagnostics: silhouette, ARI, and NMI across seeds and controls.

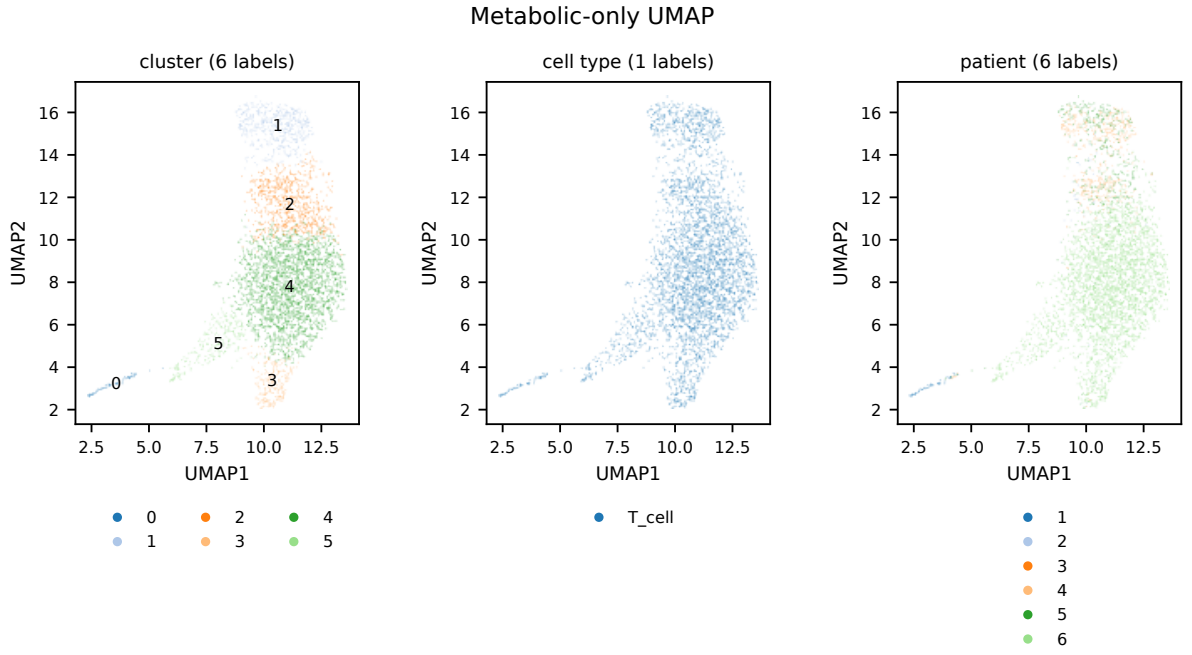


Figure 3: CD8 T cell metabolic UMAP of 3,624 cells clustered on 1,318 metabolic genes. 6 Leiden clusters at resolution 0.4.

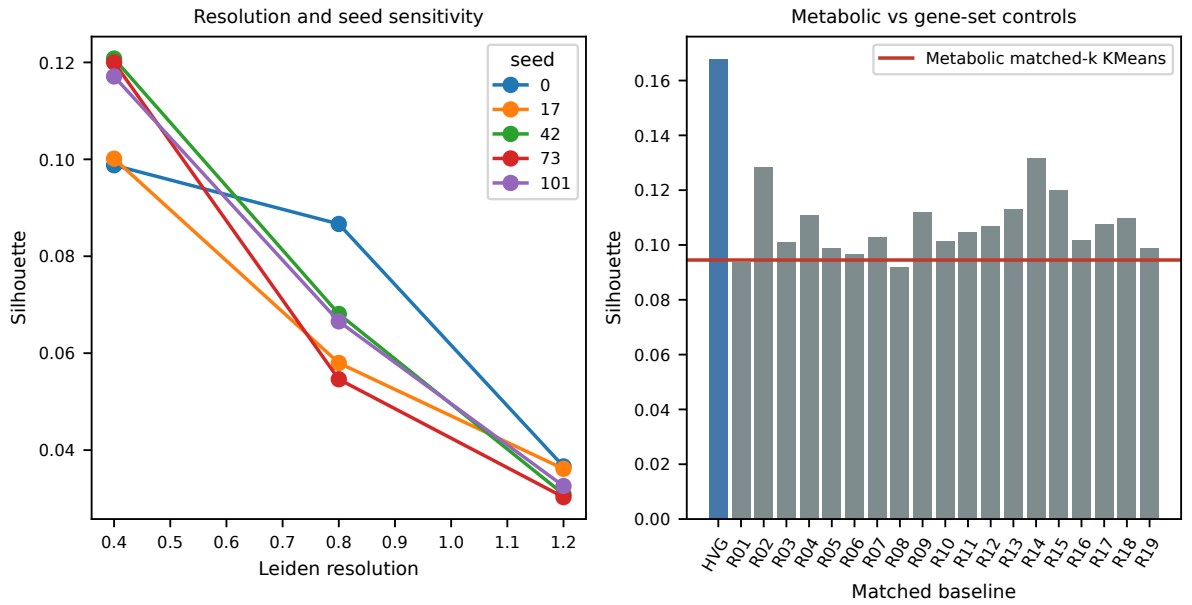


Figure 4: CD8 T cell clustering diagnostics: silhouette, ARI, and NMI across seeds and controls.